

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Operating Systems COURSE CODE: CSA04

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

INDUSTRY PROBLEM SOLVING TASK

Code of Conduct:

I _Narmatha J (192521341)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Narmatha J

Annexure A

INDUSTRY PROBLEM SOLVING TASK

S.No	Industry Problem Solving Task Questions	Marks
1	A cloud storage company is facing performance issues due to frequent file access delays. Analyze whether contiguous, linked, or indexed allocation would be most suitable for improving access speed and justify your choice with reasons.	5
2	A data center experiences high disk head movement while processing user requests. Suggest the most appropriate disk scheduling algorithm (FCFS, SSTF, SCAN) to reduce seektime and explain how it improves performance in this scenario.	5
3	An operating system uses a bitmap for free space management but struggles with slow allocation when the disk size increases. Propose an optimization technique to improve the efficiency of bitmap-based allocation and justify your solution.	5
4	A real-time application requires consistent and predictable disk access times. Evaluate whether SCAN or C-SCAN would be more suitable and explain your reasoning based on system requirements.	5
5	A company wants to design a hybrid storage system combining file allocation and disk scheduling techniques. Suggest a suitable combination (allocation + scheduling) for handling large-scale data access efficiently and justify your choice based on performance factors.	5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

ANSWERS:

INDUSTRY PROBLEM-SOLVING TASK

1. FILE ALLOCATION METHOD – CONTIGUOUS ALLOCATION

Industry Problem:

A cloud storage company is facing performance issues because users frequently access stored files. Delays in locating and reading file blocks are affecting the overall storage performance.

Analysis:

The three common file allocation methods are Contiguous, Linked, and Indexed Allocation. Contiguous allocation stores all blocks of a file in consecutive disk locations, which reduces the time needed to access the file.

Solution:

The company can use Contiguous Allocation for frequently accessed files. Since the blocks are stored continuously, the disk can access them quickly with less movement.

Performance Improvement:

- Provides fast sequential file access.
- Supports efficient direct access.
- Reduces disk head movement.
- Avoids pointer traversal required in linked allocation. .

Limitation:

It may cause external fragmentation, especially when files are frequently created, deleted, or resized.

Conclusion:

Therefore, Contiguous Allocation is the most suitable choice for this cloud storage company because it provides faster file access and improves overall storage performance.

2. DISK SCHEDULING ALGORITHM – SSTF

Industry Problem:

A data center receives a large number of user requests, causing high disk head movement. This increases seek time and results in slower data access.

Analysis:

The available algorithms are FCFS, SSTF, and SCAN. FCFS processes requests in arrival order, which may cause unnecessary head movement. SSTF selects the request closest to the current disk-head position, while SCAN services requests while moving in a particular direction.

Solution:

The data center can use SSTF (Shortest Seek Time First). It processes the nearest pending request first, reducing the distance travelled by the disk head.

Performance Improvement:

- Reduces disk head movement.
- Decreases average seek time.
- Improves response time.
- Processes nearby requests faster.
- Improves overall disk performance and throughput.

Limitation:

Requests located far from the current disk-head position may experience longer waiting time.

Conclusion:

Therefore, SSTF is an appropriate solution because it minimizes unnecessary disk head movement and reduces seek time in the data center.

3. BITMAP FREE-SPACE MANAGEMENT OPTIMIZATION

Industry Problem:

The operating system uses a bitmap to manage free disk space. As the disk size increases, the bitmap becomes very large, making the search for free blocks slower.

Analysis:

In a basic bitmap, each disk block is represented by a bit indicating whether it is free or occupied. When the disk becomes large, scanning the complete bitmap takes more time and affects allocation performance.

Solution:

A Hierarchical or Grouped Bitmap can be implemented. The bitmap can be divided into smaller groups, and summary information can be maintained for each group.

Performance Improvement:

- Reduces the amount of bitmap that must be scanned.
- Quickly identifies groups containing free blocks.
- Reduces free-space search time.
- Improves disk block allocation speed.
- Works efficiently for large-capacity storage systems.

Limitation:

Additional summary information requires some memory and management overhead.

Conclusion:

Therefore, Hierarchical Bitmap is a suitable optimization because it reduces the search area and improves free-space allocation efficiency as disk size increases.

4. SCAN VS C-SCAN FOR REAL-TIME APPLICATIONS

Industry Problem:

A real-time application requires disk requests to be processed with consistent and predictable access times. Large variations in waiting time can affect application performance.

Analysis:

Both SCAN and C-SCAN move the disk head systematically. SCAN services requests in both directions. C-SCAN services requests in only one direction and returns to the beginning after reaching the end.

Solution:

The application should use C-SCAN (Circular SCAN) because it provides more uniform waiting time for different disk requests.

Performance Improvement:

- Provides predictable disk access.
- Gives more uniform waiting time.
- Reduces variation in request response time.
- Provides systematic request processing.
- Suitable for systems requiring consistent performance.

Limitation:

The return movement of the disk head does not service requests, which can result in some additional movement.

Conclusion:

Therefore, C-SCAN is more suitable than SCAN for real-time applications because it provides more consistent and predictable disk access times.

5. HYBRID STORAGE SYSTEM – CONTIGUOUS ALLOCATION + C-SCAN

Industry Problem:

A company needs to design a storage system capable of handling large-scale data access efficiently. It requires fast file access as well as efficient processing of a large number of disk requests.

Analysis:

File allocation determines how data blocks are stored, while disk scheduling determines the order in which disk requests are processed. Therefore, selecting suitable techniques for both is important for overall system performance.

Solution:

The company can combine Contiguous File Allocation + C-SCAN Disk Scheduling.

Contiguous Allocation:

- Stores file blocks in consecutive disk locations.
- Provides fast sequential and direct file access.
- Reduces time required to locate file blocks.
- Suitable for large and frequently accessed files.

C-SCAN Scheduling:

- Processes requests in one direction.
- Provides more uniform waiting time.
- Reduces unpredictable variations in response time.
- Handles multiple disk requests systematically.

Expected Performance:

- Faster file access.
- Reduced seek time.
- Better disk utilization.

- More predictable response time.
- Efficient handling of large-scale data requests.

Conclusion:

Therefore, Contiguous Allocation + C-SCAN is a suitable hybrid solution because it combines fast file access with predictable disk scheduling, making it effective for large-scale industry storage systems.